Level 1: Users

1. Display the current username.

```
ubuntu@ip-172-31-4-201:~$ whoami
ubuntu
```

2. Create a user named `jagu`.
3. Create a user named `lilly`.
4. Create a user named `afshaan`.
5. Create a user named `soumya`.
6. Display the user ID and group ID of `jagu`.

```
ubuntu@ip-172-31-4-201:~$ sudo useradd -m jagu
ubuntu@ip-172-31-4-201:~$ whoami
ubuntu
ubuntu@ip-172-31-4-201:~$ sudo useradd -m lilly
ubuntu@ip-172-31-4-201:~$ sudo useradd -m afshaan
ubuntu@ip-172-31-4-201:~$ sudo useradd -m soumya
ubuntu@ip-172-31-4-201:~$ id jagu
uid=1001(jagu) gid=1001(jagu) groups=1001(jagu)
```

7. Check the groups of the current user.
8. Change the login shell of `lilly` to Bash using `usermod`.

```
ubuntu@ip-172-31-4-201:~$ sudo usermod -s /bin/bash lilly
usermod: no changes
```

Level 2: Groups

9. Create a group named `developers`.
10. Create a group named `testers`.
11. Add `jagu` to the `developers` group.
12. Add `lilly` to the `testers` group.
13. Add `afshaan` to the `developers` group.
14. Add `soumya` to the `testers` group.
15. Add `jagu` to the `testers` group without removing existing supplementary groups.
16. Display the groups of `jagu`.
17. Change the primary group of `lilly` to `developers`.

```
ubuntu@ip-172-31-4-201:~$ sudo groupadd developers
ubuntu@ip-172-31-4-201:~$ sudo groupadd testers
ubuntu@ip-172-31-4-201:~$ sudo usermod -aG developers jagu
ubuntu@ip-172-31-4-201:~$ sudo usermod -aG testers lilly
ubuntu@ip-172-31-4-201:~$ sudo usermod -aG developers afshaan
ubuntu@ip-172-31-4-201:~$ sudo usermod -aG testers soumya
ubuntu@ip-172-31-4-201:~$ sudo usermod -aG testers jagu
ubuntu@ip-172-31-4-201:~$ groups jagu
jagu : jagu developers testers
ubuntu@ip-172-31-4-201:~$ sudo usermod -g developers lilly
```

Level 3: File Ownership

18. Create a directory named `LinuxLab`.
19. Create a file named `test.txt` inside it.
20. Change the owner of `test.txt` to `jagu`.
21. Change the group owner of `test.txt` to `developers`.
22. Display the file's owner and group.

```
ubuntu@ip-172-31-4-201:~$ mkdir -p LinuxLab
ubuntu@ip-172-31-4-201:~$ touch LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chown jagu LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chgrp developers LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ ls -l LinuxLab/test.txt
-rw-rw-r-- 1 jagu developers 0 Oct 12 12:26 LinuxLab/test.txt
```

Level 4: Symbolic Permissions (`chmod`)

23. Give the owner read, write, and execute permissions.
24. Add write permission to the group.
25. Add write permission to others.
26. Remove write permission from others.
27. Give the group read-only permission.
28. Set owner permissions to `rwx`, group permissions to none, and others permissions to none.
29. Set permissions to `u=rwx,g=rx,o=r`.

```
ubuntu@ip-172-31-4-201:~$ sudo chmod u+rwx LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod g+w LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod o+w LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod o-w LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod g=r LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod u=rwx,g=,o= LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod u=rwx,g=rx,o=r LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ id
uid=1000(ubuntu) gid=1000(ubuntu) groups=1000(ubuntu),4(adm),24(cdrom),27(sudo),30(dip),102(lxd)
```

Level 5: Numeric Permissions (`chmod`)

30. Set file permissions to `777`.
31. Set file permissions to `755`.
32. Set file permissions to `644`.
33. Set file permissions to `700`.
34. Set file permissions to `750`.
35. Set file permissions to `640`.

```
ubuntu@ip-172-31-4-201:~$ sudo chmod 777 LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod 755 LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod 644 LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod 700 LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod 750 LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$ sudo chmod 640 LinuxLab/test.txt
ubuntu@ip-172-31-4-201:~$
```

36. Convert `rwxr-xr-x` into numeric permissions. 755
37. Convert `rw-r--r--` into numeric permissions. 644
38. Convert `rwxr-x---` into numeric permissions. 750